

How do IT service vendors build organizational resilience in response to major exogenous shocks?

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ABSTRACT

Organizational resilience is crucial for information technology (IT) service vendors, as they are highly vulnerable to major exogenous shocks. Extant resilience studies in the information systems (IS) field focus on the resilience of IS or the resilience of organizations that utilize IS. However, neither perspective adequately addresses how IT service vendors build organizational resilience in response to exogenous shocks. Using the feedback loop, this study investigates how IT service vendors interact with their environment to build resilience. Through an in-depth case study of two Chinese IT offshoring vendors, we found that IT service vendors can build organizational resilience through outside-in and inside-out processes to form a new stable state after experiencing major exogenous shocks: (1) In the outside-in process, vendors absorb external resources to build new capabilities. The combination of developing resilient IT portfolios with the collective interpretation of business domains facilitates the process. (2) In the inside-out process, vendors establish and improve client relationships by demonstrating their solutions and experiences. Our findings unpack the resilience development process, enrich the understanding of resilience in IS research, and extend the literature on IT service vendors' risk management and organizational resilience.

1. Introduction

Major exogenous shocks are unpredictable and high-risk events that seriously disturb businesses across industries and societies in a relatively short period [1] with long-term impacts [2]. These events, such as natural disasters, the financial crisis of 2008, and the recent global pandemic, create significant uncertainties and challenges for organizations, threatening their survival [3].

This research explores how information technology (IT) service vendors, offering IT-enabled applications and solutions, business processes, and infrastructure services [4], respond to significant exogenous shocks. Global IT service expenditures are projected to reach \$1.5 trillion in 2024, reflecting an 8.9 % increase from 2023.¹ Despite the substantial market size, IT service vendors—particularly small and medium-sized enterprises—remain vulnerable to demand fluctuations caused by major exogenous shocks. There are two reasons. First, clients

may view IT services as dispensable and thus postpone or even cancel related expenditures during adverse times [5]. For example, during the 2008 financial crisis, IT spending in Japan fell by 9.3 % [6], reversing the increasing trends from 2002 to 2007 [7]. Recent surveys by IDC indicated that 63 % of CIOs globally considered reducing IT expenditure after the COVID-19 pandemic.² Second, clients and vendors in IT outsourcing have low trust in each other [8], leading clients to engage in multi-sourcing [9–11] or use sequential contracts [12] with IT service vendors to mitigate risks from their opportunistic behaviors [9]. However, these contracts amplify IT service vendors' demand uncertainty during adversity [13]. All these factors make them more vulnerable to exogenous shocks [5,14].

Recent research highlights that building resilience is a crucial strategy for organizations to survive and thrive in the face of major exogenous shocks [2,15]. This is especially relevant for IT service vendors, who are particularly vulnerable to such disruptions. Organizational

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¹ <https://www.gartner.com/en/documents/5034232>.

² <https://blogs.idc.com/2022/07/22/the-first-potential-recession-in-the-as-a-service-technology-world/>.

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resilience refers to the process by which firms bounce back and bounce forward from external shocks, leading to a new stable state that aligns with the changing environment [15–19]. Resilience depends on capabilities and resources that varies across industries [20]; thus, resilience in the IT service industry is, to some extent, unique compared with other sectors [21].

However, the current research on resilience in the information systems (IS) field has overlooked the resilience of IT service vendors. Prior literature has focused on the resilience of IS (e.g., [22,23]) or the resilience of organizations that utilize IS or digital technologies to deal with major shocks and adapt to disruptions (e.g., [2,24–27]). The former stream, from the IS design's perspective, considers resilience as a property of IS, focusing on designing resilient IS [22], whereas the latter takes the perspective of IS users (e.g., IT service clients),³ viewing IS or digital technologies as tools to achieve organizational resilience by altering their routines and processes [22].

We contend that extant research addresses the resilience of IT service vendors inadequately. Compared with IT service clients, digital technologies (or IS) for IT service vendors are not just tools but an integral part of their business. This situation makes it more challenging for IT service vendors to build organizational resilience. On one hand, unlike IT service clients who adopt digital technologies as tools to alter their own routines, processes, and practices in response to exogenous shocks (e.g., [2]), IT service vendors should design and deliver resilient IS tailored to the distinct and varied clients' needs. This creates different, and often more complex, challenges for IT service vendors than for clients. On the other hand, given their vulnerability to exogenous shocks and the complex, volatile environments driven by rapidly evolving digital technologies, IT service vendors must implement coordinated management strategies and resource allocation initiatives continually to build new IT capabilities. These capabilities are essential for both bouncing back and bouncing forward [28,29], as IT capabilities form the foundation of their operations [30,31]. These dual requirements make the resilience-building process for IT service vendors more complex and underexplored. This study explores *how IT service vendors build organizational resilience to cope with major exogenous shocks*.

We draw on the resilience feedback loop framework to address this question. The framework posits that resilience develops through ongoing interactions between organizations and their environments [15, 18]. However, prior research has primarily focused on how organizations acquire external knowledge and information through learning and environmental interpretation [15], while largely overlooking how they actively shape their environments—an essential component of the resilience feedback loop. This limitation constrains our understanding of how IT service vendors build resilience. These vendors should not only absorb external knowledge [30,31] but also enhance their competitive positioning and strengthen client relationships by offering updated solutions embedded with emerging digital technologies to survive and thrive after exogenous shocks (e.g., [32]). Thus, the IT service vendor context provides a valuable opportunity to address this gap and advance knowledge on organizational resilience.

Using an in-depth case study approach, we examine how two Chinese IT outsourcing vendors responded to the 2008 global financial crisis and its aftermath from 2008 to 2015. This crisis had a huge impact on the IT outsourcing industry, resulting in a considerable shift in market demands [5], making it a suitable context for studying organizational resilience (e.g., [33]).

Our research has three potential theoretical contributions. First, we expand research on resilience in the field of IS to cover a more complex situation of IT service vendors. This opens up a new research topic for resilience in IS. Second, we explore the mechanism for IT service vendors to bounce forward to realize transformative change facing external

shocks. This goes beyond the existing risk management literature which only focuses on the reactive risk absorption and adaptation initiatives. Third, this research categorizes the resilience feedback loop into outside-in and inside-out processes, identifying ability demonstration—alongside learning and environmental interpretation—as a key activity that closes the loop between the organization and its environment. This addresses a critical gap in extant literature, which predominantly focuses on firms' exploitation of external resources (i.e., outside-in) while neglecting their influence on environmental dynamics (i.e., inside-out).

2. Theoretical foundations

2.1. Resilience in IS research

Organizational resilience describes how organizations bounce back and bounce forward from external shocks [34]. Resilience has been studied in multiple disciplines, such as engineering, management, education, and ecology [15]. In recent years, scholars in the IS field have also delved into organizational resilience, focusing on two streams: resilient information systems and resilient social systems.

The first stream of research focuses on exploring what characteristics an information system should possess to better resist external shocks and maintain system stability and functionality [22,35,36]. Properties such as flexibility [37], redundancy [23], smartness [38], and modularity [36] have been emphasized. To obtain these attributes and enhance the resilience of information systems, mechanisms such as distributed architecture, redundant design [39], security measures, and backup and recovery strategies [40] have been investigated.

The second stream regards IS or digital technologies as antecedents or enablers of organizational resilience (e.g., [2,27,41–47]), thus giving rise to the concept of digital resilience. Digital resilience is defined as the capacity of organizations to recover from exogenous shocks and even transform into new states through the design, deployment, and use of digital technologies [2,21,48]. Focusing on this point, scholars believe that firms can leverage digital technologies to adapt their internal routines, processes, and practices to bounce back and bounce forward from external shocks [2]. Specifically, IS facilitates the acquisition, integration, and application of information and resources after exogenous shocks [42,49,50]. For example, the intelligent perception characteristics of digital technologies (such as real-time analysis and intelligent algorithms) assist entities in collecting and analyzing data to predict and respond to shocks [2]. In addition, IS enhances organizational resilience by promoting information and knowledge sharing and collaboration within an organization during exogenous shocks [25,51]. Knowledge sharing and collaboration lead to joint problem-solving activities across the firm to adapt to new environments [33,52]. Furthermore, IS enhance organizations' innovation in new business models, products, or service [2,53], which can improve performance and help them transform into a new stable state after exogenous shocks [54]. For example, Nielsen et al. [27] explored the dynamic resilience-building process and the role of digital technology when a university shifted to virtual teaching amid a government-mandated lockdown during the COVID-19 pandemic.

These two streams are from the perspective of either the IS design or the IS users (i.e., IT service clients). Since resilience is intrinsically linked to an organization's core capabilities and resources [20], the resilience of IT service vendors who provide IT services and have different core capabilities with IT service clients has been largely ignored. Unlike IT service clients, who leverage digital technologies to adapt their internal routines and processes for recovery and progression [2], IT service vendors must design information systems tailored to the distinct and varied needs of their clients. In other words, while digital technologies serve as instruments for clients, they represent the fundamental capabilities of IT service vendors. In addition, as digital technologies are evolving rapidly, and so is the market demand brought by emerging digital technologies, IT service vendors are not only required to offer

³ In this research, IS users refers to organizations that apply IS or digital technologies in their operations, particularly IT service clients.

customers continuously updated and more resilient information systems and services in terms of business output, but also need to be flexible in their business processes to respond to changes more rapidly. This exerts higher requirements on IT service vendors to bounce back and forward swiftly from exogenous shocks. Therefore, it is necessary to discuss the resilience of IT service vendors further.

2.2. Organizational resilience of IT service vendors

Organizational resilience is essential for organizations to respond effectively to adversity [15]. In essence, resilience serves as a strategy for navigating environmental uncertainties and risks [16] and is interrelated with risk management to some extent [15]. The literature on risk management discusses how IT service vendors respond to exogenous risks, offering valuable insights into the resilience of IT service vendors.

IT service vendors are exposed to various exogenous risks, such as financial crises [55,56], natural disasters [57]; [13,58–60], and demand uncertainties [5]. A few studies have investigated how IT service vendors respond to exogenous shocks. These research suggests that IT service vendors can manage risks by developing intra-industry partnerships to facilitate risk-sharing [61], establishing contracts and service portfolios to control risks [55,62], and enhancing their operational capabilities, such as client-specific and software development capabilities, to mitigate the negative impacts of exogenous risks [5,63,64]. For instance, some studies indicate that IT service vendors align their processes with those of their clients [65,66] or invest dedicated resources to serve specific clients [67]. These strategies can enhance client trust [68,69] and improve client-specific capabilities, ultimately strengthening collaboration with clients and reducing demand uncertainties [64,70].

Although these measures can to some extent promote the bounce back of IT service vendors, resilience requires organizations not only to react to crises, but more importantly, to achieve transformative changes to better cope with future environmental changes [15]. In other words, they need to form a proactive response to crises and bounce forward to develop resilience. Especially for IT service vendors, major disruptions disrupt the whole business environment, and they often have to turn to adjacent trajectories [32,71], which may involve shifting to new industries [5] or moving upstream in the value chain [72]. This transformation requires the development of new IT services and capabilities [72]. However, previous measures and capabilities may even prevent companies from responding proactively to shocks. For example, original client-specific capabilities and the specificity of IT assets may not fit new market contexts and can even hinder vendors' ability to bounce forward [32]. Therefore, further discussion is needed to understand how IT service vendors can proactively deal with change and build forward better from shocks and crises through resilience.

2.3. Resilience feedback loop

Organizational resilience develops over time after exogenous shocks [16–18,73]. It involves continuously acquiring, building, integrating, and allocating resources and capabilities to absorb the adversity, adapt to new environment (i.e., bounce back), and ultimately transform to a new and stable state aligned with the new environment (i.e., bounce forward) [3,27,74]. This process reflects the dynamic interplay between organizations and their environment [71,75]. Organizations not only acquire and integrate resources from their environments but also shape the very environments in which they operate. These environmental changes, in turn, exert a reciprocal impact on organizational actions [18,52]. Through their actions and the integration of feedback from these actions, organizations contribute to the continuous development of resilience [15]. This iterative process of building organizational resilience through dynamic interactions between an organization and the environment is conceptualized as the resilience feedback loop [15].

Key actions within the resilience feedback loop include environmental interpretation and learning. Environmental interpretation refers

to managers analyzing and assessing future challenges, disruptions, and potential opportunities [15,76,77], which decides how organizations take appropriate tasks and shape responses to adversity [15,74], such as developing new businesses (e.g., [54]), restructuring organizational structures and culture (e.g., [43]), or leveraging digital technologies [2]. Resources such as data, information, and knowledge acquired from external sources help organizations interpret new environments [19,50]. For instance, during the COVID pandemic, hospitals and governments utilized real-time dashboard data to anticipate the progression of the pandemic and formulate plans for resource allocation [2].

Learning helps organizations build new capabilities and adapt to the environment [50]. Organizations learn from experiences with adversity by encoding and recombining new knowledge gained from initial responses [18,74] to redesign their businesses and build new capabilities. For example, organizations understand new requirements and redesign their businesses to meet requirements during their cooperation with clients [45,78]. Learning from experience further influences organizations' interpretation of new environments and their subsequent responses [15]. As a result, organizations can achieve a new equilibrium with the new environment [75].

Environmental interpretation and learning primarily describe organizational actions of acquiring information and knowledge from the environment, but the extant literature largely overlooks how organizations actively influence their environments, which is a critical component for closing the resilience feedback loop. This research gap undermines the framework's ability to explain how IT service vendors build organizational resilience. To achieve resilience, IT service vendors must not only acquire external knowledge necessary for capability development [30,31] but also proactively shape their environment conditions through high-value service offerings (e.g., [32]). For example, IT service vendors enhance their competitive positioning within the industry and strengthen client relationships by delivering updated products and services embedded with emerging digital technologies [79]. Investigating the resilience-building process of IT service vendors thus addresses this gap and advances theoretical understanding of the resilience feedback loop.

3. Research method

We conducted an in-depth case study for the following two reasons. First, case study research is an ideal approach for understanding complicated processes [80,81] and exploring "how" questions [82]. Since our research focuses on how IT service vendors build resilience, which is a sophisticated and complex change process [83], case study research is appropriate. Second, the case study is also recommended for investigating organizational resilience [75,84].

Specifically, our case study used a matched pair design, which involves comparing two cases with similar initial states but different subsequent processes and outcomes [85]. This design helps in identifying contrasting patterns and uncovering the underlying causes of the patterns [80]. In this research, one of the cases becomes resilient via transformation, while the other is unable to cope with the environmental shock. This contrast can shed light on the process of building organizational resilience.

We selected two IT outsourcing vendors, Beta and Alpha, as our sample for the following reasons. First, Chinese IT outsourcing vendors experienced a radical transition in coping with the 2008 global financial crisis, which was one of the most critical events of modern socioeconomic history [86]. This context provided an opportunity to understand organizational resilience in times of crisis [15]. Previously, the IT outsourcing industry experienced phenomenal growth by focusing on

niche overseas markets, which left them lacking the capabilities and resources to adapt to other markets [87]. Nevertheless, in response to the financial crisis and a rapidly shrinking overseas market, most vendors attempted to shift from offshore markets to the booming domestic market.⁴ They tried to provide IT products and solutions for domestic clients instead of coding and testing for overseas IT companies, which required new capabilities of these vendors. In addition, after the financial crisis, IT service vendors experienced a period of rapid technological development (such as big data, the mobile Internet, and the Internet of Things). New IT services, such as e-commerce solutions, gained momentum with these new technologies, creating new opportunities and disruptions for the IT service industries. Since building organizational resilience is a long journey of adaptation and transformation for firms [88], this research context is well suited for studying resilience (e.g., [33]).

Second, Beta and Alpha had a similar initial state, experienced the same environmental changes, and made the same strategic change, but the outcomes were vastly different, forming a sharp contrast (see Table 1, Figs. 1 and 2). Alpha and Beta both relied heavily on offshore outsourcing for over 90 % of their revenue before 2008. However, after the sharp decline in their overseas orders in 2008, they both downsized their operations and began exploring the domestic market in 2010. By 2015, Alpha had shrunk to just 60 employees, focusing on providing custom development services for domestic clients by leveraging its original core capabilities. On the other hand, Beta not only successfully developed new capabilities and transformed itself into an enterprise Internet ecosystem solution provider but also recovered and surpassed its historical peak in organizational scale.

3.1. Data collection

This research is part of a long-term program on IT outsourcing vendors that has been ongoing since 2004. Multiple rounds of interviews were conducted at Alpha in 2004, 2007, 2010, and between 2012 and

Table 1
Basic information of sample companies.

	Alpha (locked in the existing path)	Beta (path transformation)
Year of foundation	2001	2004
Primary market in 2008	Japan	Europe and the U.S.
Primary markets in 2015	China (profit: 70 %; employees: 30 %) Japan (profit: 30 %; employees: 70 %)	Europe and the U.S. (sales: 50 %) China (sales: 50 %)
Size in 2008	Over 240 employees	Over 200 employees
Smallest size from 2008 to 2015	About 60 employees	About 70 employees
Size in 2015	About 60 employees ^a	More than 200 employees
Main businesses by 2008	Coding and testing for large Japanese software companies	Coding and testing for large European and U.S. software companies
Main businesses in 2015	Coding and testing for Japanese software companies; custom development for Chinese firms	Enterprise Internet ecosystem solutions such as e-Commerce systems, supply chain solutions, and big data analysis

^a In 2016, Alpha was acquired by a consulting company and became its wholly owned subsidiary.

⁴ These changes were consistent with the *China Software and Information Service Outsourcing Industry Development Report 2010*, published by the Ministry of Industry and Information Technology of China.

2015. We initially focused on capability building, and later on vendors' response to the financial crisis and their further development. In November 2014 and May 2015, we conducted additional investigations on Beta, which successfully entered the domestic market. The first and second authors conducted all interviews, while the third author participated in data collection in 2014 and 2015. We did not analyze the interview data in 2004 and 2007 because they were irrelevant to the financial crisis, and thus they are not listed in Table 2.

Semi-structured interviews were conducted at each company with at least one corporate-level executive, department heads, and project managers who had worked for the company for at least five years (see Table 2). The interviews were conducted in a top-down sequence (e.g., [66]). In each company, an executive was interviewed first, who was then asked to refer researchers to several business department heads. We focused on how IT vendors recover from exogenous shocks; therefore, we opted to interview department heads who had attempted to explore new products or clients in both domestic markets and offshoring outsourcing markets. In turn, they were asked to recommend experienced project managers and developers who had participated in market exploration—such as conducting domestic outsourcing projects or developing new products—for interviews. Using this top-down mandate method helps secure the participation of key informants with expertise in market exploration and improves the quality of interview data [89]. For each round of data collection for each company, we ended the current interview once no new information emerged. During data analysis, we communicated with informants via email, phone, or social media to confirm details, seek clarification, supplement missing information, or resolve inconsistencies.

Most interviews lasted from 60 to 90 minutes. Key interview questions at interviews after 2008 included the following: What impact did the financial crisis have on your company? How did you and your organization/department/team respond to it? What were the differences between the domestic market and overseas markets? To serve domestic clients, what measures did you and the firm take? What new developments has the firm experienced since the last interview? Most key questions were answered by multiple interviewees in each firm for triangulation.

Furthermore, we collected secondary data to supplement the primary data collected (refer to Table 3 for details). First, we regularly monitored the websites of both companies to keep track of their latest developments, although Alpha ceased updating its websites in 2011. The information collected from the website corroborated and supplemented the interview data, particularly in verifying the timing of key events. Second, we reviewed seven China IT outsourcing industry reports published by Chinese governments, which offered insights into the market environments and development trends of the IT outsourcing industry.

3.2. Inductive data analysis

This in-depth case study employs inductive analysis to build theory from data. Inductive analysis approach facilitates revealing complex processes [80] by reasoning from specific data in local contexts to the theory [90,91]. Our analytical approach comprises four iterative phases, each containing multiple steps, as detailed in Table 4. These steps are iteratively applied throughout the whole process.

Phase I: In within-case analysis [92], we collated the interview data and second-hand information to clarify the developmental trajectories of the two cases from their foundations to 2015. The focus was on the processes to identify critical events that influenced subsequent development and response actions. Then we used temporal bracketing [93] to divide each case into two stages labeled as responding to adversity and forming resilience feedback loops (see Figs. 1 and 2). For Alpha, the critical juncture between the two stages was identified as the point when Partner Fu took charge of the firm after Zhang's departure. For Beta, the stage of forming a resilience feedback loop commenced with the establishment of a strategic consensus to become an e-commerce

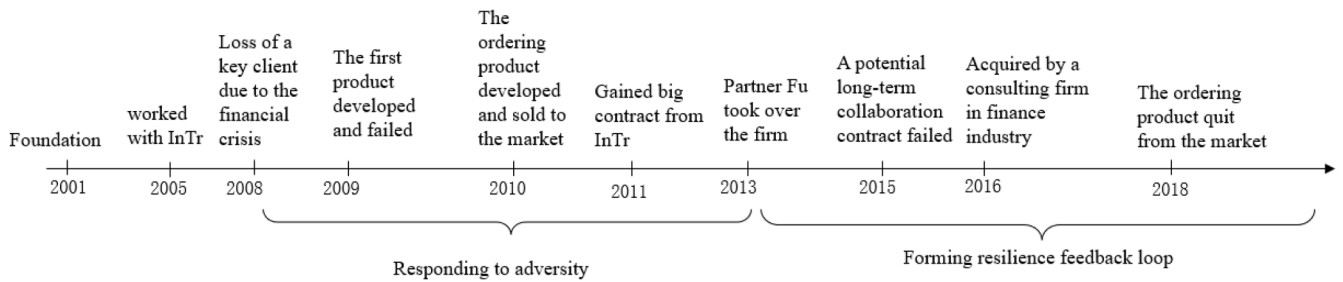


Fig. 1. Alpha's development process.

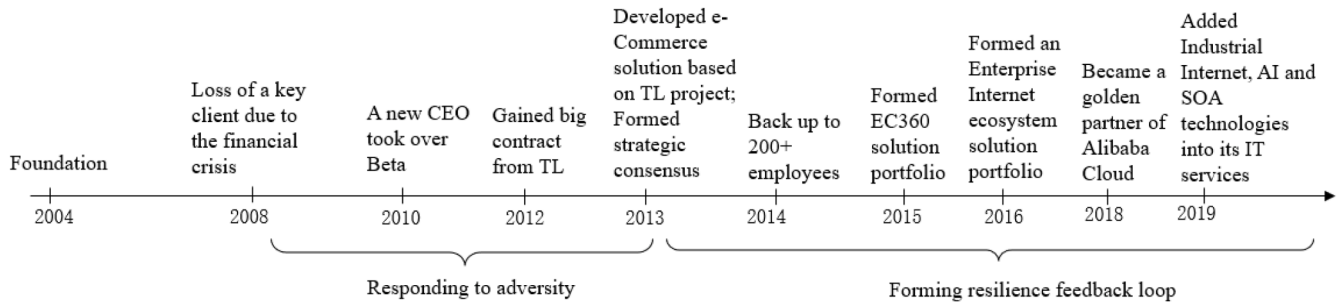


Fig. 2. Beta's development process.

Table 2
Interview details.

Case	Respondents	No. of interviews	Interview time
Alpha	General manager	2	July 2013, January 2015, and December 2018 ^b
	Corporate partners/business unit (BU) head Fu (3), Zhang (4), and Yang	9	August 2010, April 2012, July 2013, January 2015
	BU head Hao	8	August 2010, April 2012, July 2013
	Project managers of Japanese projects and domestic projects	2	July 2013
	Team leaders	2	August 2010
	Developers	23	—
	Overall	23	—
Beta	COO Ting	1	November 2014
	BU heads, Zhong, Cao(2), Li,	4	November 2014 and May 2015
	Engineering director	1	May 2015
	Project managers	2	November 2014 and May 2015
	Human resource manager	1	November 2014
	Requirements analyst	1	May 2015
	Developer of mobile Internet	1	November 2014
	Overall	11	—
	Industry experts	3	January 2015, November 2014
Total		37	—

Notes: Alpha's general manager, Mr. Fu, who used to be a partner, assumed the role in 2013 after partner Zhang's exit. Therefore, in data collected before 2013, Fu is referred to as "partner," whereas after 2013, he is designated as "general manager." Data collected before 2010 were largely irrelevant to environmental shocks and thus are sparsely used.

solution provider.

Phase II: In the cross-case analysis phase, we compared the different stages within the case and the same stage between Alpha and Beta, aided by tables and figures. Initially, the focus was on primary IT services, the market, and corresponding clients in each stage. Three types of IT services were identified in the responding-to-adversity stage: offshore IT outsourcing, domestic custom development, and IT solutions/products.

Table 3
Data sources and insights.

Sources	Descriptions	Insights
Interviews	37 in total	Provide detailed strategic activities of Alpha and Beta for us to better understand the process of building resilience
Online information	Homepages of Beta and Alpha	Provide supplementary data, such as time points of key events
China Software and Information Service Outsourcing Industry Development Report Summery (2010, 2011, 2012, 2019)	Published by Ministry of Industry and Information Technology of China	Provide information on the industry environment of Alpha and Beta, such as trends of demands in the domestic market, which helps us better understand interactions between the environment and organizations
Beijing Service Outsourcing Industry Development Report (2010, 2011, 2012)	Published by Beijing Municipal of Commerce Bureau	

At the same time, two kinds of positive feedback were identified: positive client-specific feedback and positive market feedback. The former refers to vendors' services or products being adopted by or appealing to specific clients, while the latter refers to vendors' services or products being adopted by or attractive to multiple clients in the market. Positive feedback exhibits varying degrees of strength [94]. In this research, we measured positive feedback by estimating the number of external adopters.

Phase III: We unpacked the resilience feedback loop by analyzing the raw data and paying attention to incidents and data passages that described environmental-organizational interaction actions for each case. Multiple iterations reduced to 27 first-order in vivo codes that captured firms' actions that help to recover from external shocks (see Fig. 3 and Table A1 in the Appendix). We then differentiated and clustered these first-order codes into meaningful second-order themes to describe and explain how firms interacted with external environments to build resilience. This step requires consistently comparing between data and the literature [95] and between Beta's data and Alpha's data. Nine

Table 4
Data analysis processes.

Phase	Analysis activities	Outcome
Phase I: Within-case analysis	• Describing each case's development course • Identifying critical events	• Each firm's timeline is divided into two stages: responding to adversity and forming a resilience feedback loop
Phase II: Identifying business models and corresponding positive feedback	• Identifying primary IT services, market, and clients in each case's response to adversity • Comparing the positive feedback of three IT services in each case and the capabilities required	Three types of IT services and required capabilities: • Offshoring outsourcing development: client-specific capabilities and software development capabilities • Domestic custom development (whole life cycle): consulting, sales, and software development capabilities • Providing IT products/solutions: solutions/products offering, consulting, and sales capabilities Two kinds of external positive feedback: • Positive client-specific feedback • Positive market feedback
Phase III: identifying response actions to unpack the resilience feedback loop	• Identifying managerial actions in forming a resilience feedback loop • Comparing second-order themes of Alpha and Beta by within-case similarities and between-case differences	First- and second-order codes: • 111 data passages categorized into 27 first-order codes • Then data were clustered into 9 second-order codes: 7 second-order codes emerged from Beta's data, while 5 second-order codes emerged from Alpha's data. 3 second-order codes emerged from the data of both firms. • Four high-level second-order themes were identified to characterize differences between Beta and Alpha: Developing resilient IT portfolios, including IT service differentiation/ integration/ optimization • Developing isolated IT portfolios (Alpha), including IT service differentiation/ optimization • Collective interpretation of business domains (Beta), including firm-level domain knowledge acquisition and integration, collective assessment of business domains • Individual interpretation of business domains (Alpha), including BU-level domain knowledge acquisition and integration, individual assessment of business domains

Table 4 (continued)

Phase	Analysis activities	Outcome
	• Forming aggregate dimensions	Three aggregated dimensions: • IT service adaptation: Developing resilient IT portfolios (Beta) vs. developing isolated IT portfolios (Alpha) • Environmental interpretation: Collective interpretation of business domains (Beta) vs. Individual interpretation of business domains (Alpha) • Ability demonstration: solution and experience demonstration (Beta) vs. experience demonstration (Alpha)
Phase IV: Integrating findings to build a theory	• Macro-level comparative analysis of Alpha and Beta • Focused on relationship between managerial actions, external positive feedback, and strategic patterns	A theory of building organizational resilience of IT service vendors (Fig. 6)

meaningful second-order themes emerged from the data, including IT service optimization/differentiation, firm-level/BU-level domain knowledge acquisition and integration, individual/collective assessment of business domains, and solution/experience demonstration.

Cross-case analysis revealed differences and similarities in building resilience between Beta and Alpha. For example, Beta conducted IT service optimization, differentiation, and integration, while Alpha only conducted IT service optimization and differentiation. We categorized them into two dimensions, developing resilient IT portfolios and developing isolated IT portfolios. We also found that Beta undertook domain knowledge acquisition/integration and business domain assessment at the firm level, while Alpha conducted these activities at the BU level. We aggregated Beta's two second-order themes (firm-level domain knowledge acquisition and integration; collective assessment of business domains) into collective interpretation of domain businesses. In contrast, Alpha's two second-order themes (BU-level domain knowledge acquisition and integration, individual assessment of business domains) were integrated into individual interpretation of business domains. Aggregating these second-order themes enabled a parsimonious comparison between the two cases.

Then we focused on similarities on second-order concepts and grouped nine second-order themes into three aggregated dimensions—IT service adaptation, environmental interpretation, and ability demonstration—based on how the flow of resources and capabilities is directed, without considering differences between Beta and Alpha but focused on similarities on second-order concepts. For instance, IT service adaptation and environmental interpretation demonstrate how companies acquire and integrate external resources to alter their business and understanding, whereas ability demonstration shows how companies change their external environments. Fig. 3 illustrates the emerging code structure (see the middle part) and the differences between Beta and Alpha through cross-case comparison.

Phase IV: To integrate findings into a theory, we performed a macro-comparison of Alpha and Beta to identify the process of building

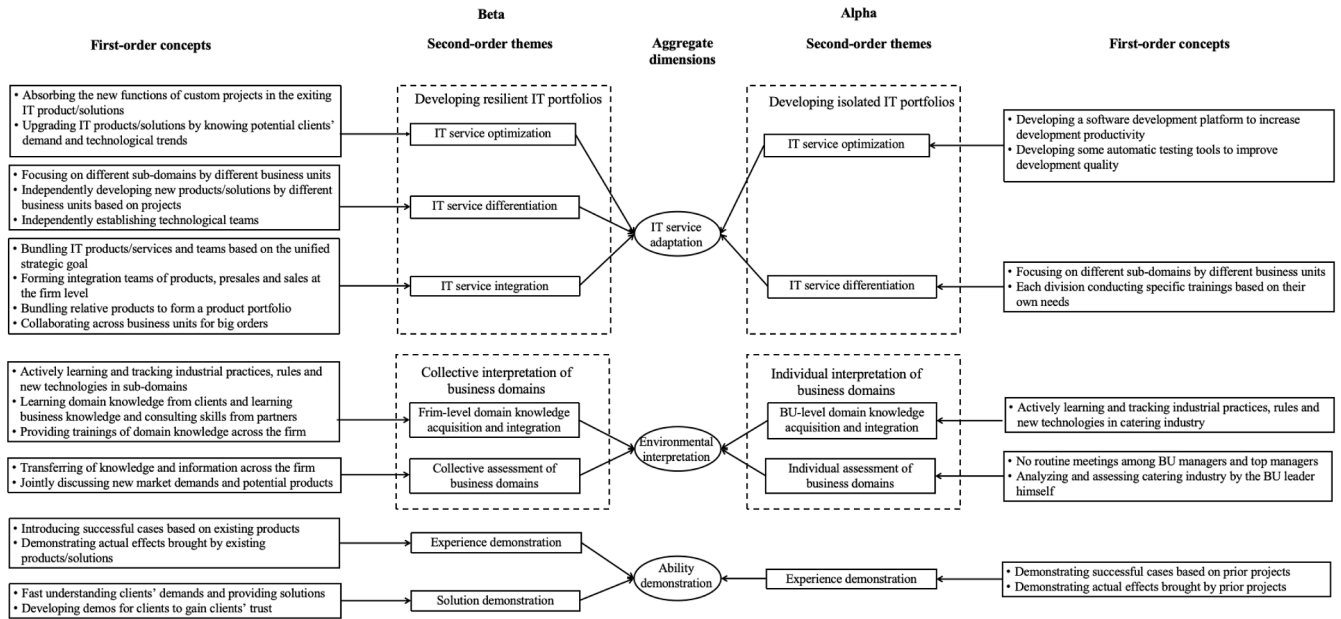


Fig. 3. Data structure.

organizational resilience regarding IT business, actions, and positive feedback, and to clarify their relationships in building organizational resilience. We also shared the main findings with key informants for validation in the final analysis phase.⁵

4. Research findings

In this section, we describe the initial states of the two cases and their following exploration in the domestic market. Then we present their resilience feedback loops in Sections 4.3 and 4.4, respectively, according to the results of phases II and III of data analysis.

4.1. Initial states

Alpha and Beta experienced rapid growth, employing over 200 people within 3–4 years of establishment. Their primary businesses were providing software development, maintenance, and testing for overseas software companies, instead of end users, in long-term collaborations. Both companies had built client-specific capabilities, that is, the capacity to adjust internal resources and routines based on a specific client's goals, communicate and collaborate effectively with the client [66]. They also obtained CMMI3 certification in 2007, a widely adopted quality assurance system for software companies, demonstrating their established software development capabilities [96].

In turn, their strong client-specific and software development capabilities led to an increase in orders from clients. Two primary clients of Alpha accounted for over 80 % of its revenue, while the biggest client of Beta contributed 40 % of its revenue. Therefore, positive client-specific feedback was evident, as two firms' services were adopted by a limited number of clients over the long term (discussed in Table 5).

4.2. Initial response to adversity: exploring alternative IT services in the domestic market

In 2009 and 2010, Alpha and Beta were negatively impacted by the

2008 financial crisis. During this period, Alpha failed to secure any new orders after losing a key client that accounted for 40 % of its revenue, except for maintenance contracts from its existing clients. Despite attempts to explore new markets, its sales efforts were fruitless. As a result, Alpha incurred a loss for three consecutive years, from 2009 to 2011. Partner Zhang recalled in 2010,

In May and June of 2008, we start noticing the reduced contract and shrinking business volume ... Starting in September, we finally realized something was going wrong because previous intended contracts were canceled. We started laying off underperformers.

Similarly, Beta also faced challenges when a key client terminated its collaboration in 2008, leading to a period of turmoil. Despite hiring three local salespeople, Beta saw no improvement. Cao, a business unit (BU) manager, recalled,

By 2008, when everyone had nothing to do, the company began to lay off employees. As soon as the financial crisis passed, only 76 people remained in our company, and then we had to terminate the lease on the third floor of this building, only renting one floor for work.

Both Alpha and Beta relied heavily on existing clients for orders, and the termination of collaboration with key clients disrupted their positive client-specific feedback.

In contrast, while overseas markets were shrinking, China's domestic market experienced explosive growth in IS demand. According to government statistics, the domestic software industry reached 1.3 trillion RMB in 2010 (approximately 195 billion USD) and 1.84 trillion RMB in 2011 (approximately 291 billion USD). The year-on-year growth rate in 2010 was 31.8 %, compared with 25.6 % in 2009. Furthermore, e-commerce in China had been on the rise since 2009 (see Fig. A1 in the Appendix). Both Alpha and Beta recognized the immense potential of the domestic market. Beta's COO, Ting, noted that many traditional firms were eager to embrace e-commerce, creating opportunities for IT services. Hence, Beta shifted its focus to the domestic market in 2009 and 2010.

There were two types of IT services in the domestic market. The first was custom development services, with contracts often secured through personal relationships (see Table 5). Domestic clients often outsource full-lifecycle IS projects to vendors, encompassing requirements analysis, system design, detailed design, coding, unit testing, integration testing, structural testing, and operation and maintenance support.

⁵ We shared our research findings with Fu and Zhang of Alpha, to answer the questions of why they failed to expand the domestic markets. However, we were not able to share our findings with Beta because we lost access to the firm due to the departure of our initial contacts.

Table 5
External positive feedback of alternative strategic options of Alpha and Beta.

Alternative strategy	External positive feedback	Description
Offshore IT outsourcing	<pre> graph LR A[More orders from this specific client] -- "+" --> B[Client-specific capabilities & Software development capabilities] B -- "+" --> A </pre>	Positive client-specific feedback Overseas clients were usually large software vendors or IT companies that subcontracted low-end software development projects to Chinese vendors. These clients place great importance on vendors' software development capabilities and client-specific capabilities. When working with a client, a vendor gradually gains an understanding of the client's system development and work patterns, which enables them to build client-specific capabilities and undertake various projects for this client. By accumulating client-specific and software development capabilities, the vendor can satisfy the client's requirements, which in turn may led clients to place more orders.
Domestic custom development (full life-cycle model)	<pre> graph LR A[More diverse orders from different clients] -- "+" --> B[Consulting/Sales capabilities & Software development capabilities] B -- "+" --> A </pre>	Positive market feedback Domestic and end-user clients need vendors' consulting and software development capabilities. By providing services to multiple clients, a vendor can accumulate domain knowledge and sales experience, which improves their consulting and sales capabilities. In addition, executing projects for diverse clients enables vendors to build software development capabilities. Developing these capabilities further enhances service quality and client satisfaction, helping the vendor secure more orders from the same industry sector. By undertaking diverse projects, vendors can continuously advance their consulting, sales, and software development capabilities.
Solution/ product	<pre> graph LR A[Adopted by more clients in the market] -- "+" --> B[Solutions/products offering capabilities, Consulting & Sales capabilities] B -- "+" --> A </pre>	Positive market feedback Domestic clients are end-user firms that value vendors' consulting and sales capabilities. Solutions/ products are customized for clients based on their specific requirements in addition to generic needs. By working with different clients, a vendor can continue to improve the quality of solutions/ products and accumulates domain knowledge and sales experience, which can further improve its consulting and sales capabilities. As product quality improves, the vendor's offerings become more competitive, attracting more orders from the market and fostering a cycle of continuous improvement in both products and capabilities.

Alpha and Beta were involved in several such projects. For example, Alpha developed an e-commerce website for an investment company, a short message system for a winery, a membership management system for a travel agency, and a B2B (business-to-business) platform for one of the largest travel agencies. Similarly, Beta conducted IS development services for a ministry of the central government, local banks, and a museum.

The second type involved providing IT products or solutions based on the vendor's experience of servicing overseas clients (see Table 5). Such solutions/products were developed based on the general practices of the industry and could meet most of the needs in the industry. Vendors required a deep understanding of the industry, which could be acquired by serving multiple clients [31]. Alpha developed an e-commerce product for small and medium-sized travel agencies based on its collaboration with InTr (a pseudonym for one of the biggest travel agencies in China) and an order processing system for the catering industry. Meanwhile, Beta developed an e-commerce solution, warehouse management systems, and supply-chain management systems based on expertise gained from custom development projects for overseas clients.

However, Alpha incurred losses from 2009 to 2011 despite small gains in the domestic market, and Beta continued to lay off employees. The main reason was that the original capabilities (the client-specific capabilities and software development capabilities) were not suitable to the new environment. To serve the domestic market, vendors needed to develop new capabilities (outlined in Table 5). First, personal relationship or guanxi is critical for marketing and gaining collaboration in the domestic market. Since Alpha and Beta's main markets were overseas before 2008, penetrating the domestic market was a significant challenge. They had to build new sales capabilities to thrive in the

domestic market.

We did not need to maintain domestic social relationships when we conducted Japanese offshoring outsourcing ... [However], to expand [the] domestic market, our weaknesses are marketing and sales capabilities. We want to improve our marketing and sales capabilities, but it requires people and money. We do not have these now because the financial crisis has exhausted us (Partner Fu, Alpha).

Sales capabilities are critical to exploring the domestic market. The practice of domestic sales is entirely different from that of the overseas market. We must recruit experienced domestic salespeople instead of training them in-house (COO Ting, Beta).

Second, consulting capabilities were necessary to serve domestic clients who lacked IT knowledge. However, when working with overseas IT companies in the past, neither Alpha nor Beta had ever conducted requirements analysis. As a result, they had acquired very little domain knowledge, which was essential for providing consulting services. Project Manager Ma (PM A1) of Alpha said in 2010,

The challenges in the domestic market are enormous. In the past, we had few chances to interact with end clients. If we undertake domestic projects, we have to do analysis requirements ourselves. This posed a significant challenge for us—providing business and IS suggestions to domestic clients.

Beta's COO Ting shared a similar view: *To serve domestic clients, we should have an industry consultant who can advise customers on what kind of system they need and how to improve their business.*"

4.3. Beta's resilience feedback loop: positive market feedback and new capabilities drive the formation of a new stable state

Through its initial responses, Beta triggered positive market feedback through different strategic options, leading to an influx of projects and the development of new capabilities. This further strengthened positive market feedback, forming a resilience feedback loop. We will discuss the loop in detail in the following section.

4.3.1. Positive market feedback generated by three alternative IT services

Beta used to serve only overseas clients (see Table 6), even during tough times between 2008 and 2010. However, despite three years of hard work, Beta was unable to receive positive market feedback from overseas markets due to sluggish growth. As a result, Beta shifted its focus to the domestic market in 2010.

In its early days as a specialized offshore IT outsourcing company, Beta only received small-scale domestic projects (with revenues ranging from US\$30,000 to 70,000) and faced difficulties such as unclear specifications, frequent changes in requirements, and challenges in receiving total payments for the services rendered.

Incidentally, a project from TL Company (one of the largest appliance manufacturers in China) presented a new opportunity for Beta. In October 2011, Beta was invited to join TL's e-commerce platform project as a partner of a leading consulting firm. Through their joint efforts, the online sales platform was launched on September 1, 2012, and achieved great success. *Thanks to this e-commerce store, online orders had been shipped three times faster than before*, according to Li, the manager of this project. Encouraged by the success, Beta further developed the e-commerce platform for generalized use and packaged it as a solution product for sale (i.e., e-DRP, electronic distribution resource planning platform).

This product promoted Beta's custom development services in the domestic market, triggering positive market feedback (i.e., Beta's e-commerce custom development services were adopted by more clients).

Table 6
External positive feedback generated by strategic options of Beta.

Potential strategic options	Positive client-specific feedback	Positive market feedback
Offshore custom development	+: No new projects; only maintenance projects from existing clients.	N/A: No new clients.
Domestic custom development	N/A	+: Custom development projects are characterized by a small scale, long cycle, and low profit margin. "We used to work for state-owned organizations such as a provincial institute of archaeology, and quickly realized that these projects would lead to sustain losses." (COO)
Solutions/Products offering	N/A	++: In two years, the e-commerce solution had served six big clients; by November 2016, its supply chain management solution had served four big clients. Big data analysis packages were adopted by four clients. The total order number of Beta showed a significant upward trend. "We explored approximately 20–30 new clients in the domestic market." (BU manager Li)
Overall	+	+++

Note: ++ refers to strong external positive feedback.
+ refers to weak external positive feedback.
N/A refers to that no feedback is found.

Based on its success with TL, e-DRP with certain degrees of customization was adopted by two other leading appliance manufacturers (order values exceeding RMB 1 million). From 2013 to 2015, Beta further developed e-commerce platforms for three additional clients, including a leading clothing manufacturer, a multinational electronic appliance manufacturer, and a national chain of laundry stores (see Table 6).

In return, the custom development projects allowed Beta to assess market potential and refine the IT solutions through multiple implementations. As COO Ting explained,

Our company will not invest a large sum of money to develop standalone software products. Instead, we build IT products by redeveloping system based on customized IS development projects. Through deploying and implementing the product for many customers, we have continuously improved it. If the product does not sell well, we just drop it and develop another one.

The successive adoptions of e-DRP, that is, positive market feedback, demonstrated the promising prospects of e-commerce and boosted their confidence in e-commerce solution. Therefore, Beta shifted its strategic focus to solutions/products, especially to e-commerce solutions, in August 2013. Cao, the business unit (BU) manager, concluded,

This product (i.e., e-DRP) has given us great confidence. We think that e-commerce may be a way for us to survive better in China, so we are committed to transforming our business.

4.3.2. Developing resilient IT portfolios and collective interpretation of business domains to build new capabilities

The increasing number of projects, that is, positive market feedback, brought Beta new domain knowledge, new technological applications of e-commerce, and increasing market demands. To meet the demands of the domestic market, Beta designed resilient IT portfolios and fostered collective interpreted to build new capabilities.

Collective interpretation of business domains. Collective interpretation of business domains is an activity in which organizational managers and members jointly analyze, assess, and identify opportunities and threats of business domains [97], including collective assessment of market demands and firm-level domain knowledge acquisition and integration. Business domains involve function domains such as e-commerce and supply chain management applications [31,66] and industry domains such as banks, museums, and the clothing industry [30].

Firm-level domain knowledge acquisition and integration is the process of learning and integrating domain knowledge at the firm level from external sources such as clients, partners, books, and the Internet. In Beta's case, all four BU managers mentioned that they learned through frequently communicating with potential clients and business partners. For example, Beta sought to stay abreast of cutting-edge industry practices (e.g., the new rules of the top e-commerce websites such as Tmall.com and JD.com) and the latest technologies, such as the latest version of the iOS or Android operating systems. In 2014, Beta established a business consulting team composed of IT engineers with a propensity for consulting. With 2–3 members dedicated to each industry, the team systematically learned, accumulated, integrated, and disseminated domain knowledge specific to their sectors, thereby enhancing the company's consulting capabilities.

Beta conducted a *collective assessment of market demands*. Its BU managers, sales, and consultants jointly assessed contextual information regarding business domains, such as project leads, market demands, and new technological trends. The sales department, BU managers, and top managers attended routine Monday morning meetings to review all project leads and progress. Furthermore, the BU managers attended routine Wednesday afternoon meetings to discuss opportunities for potential products or solutions and compare each strategic option based on market potential, profitability, and long-term viability. For example,

Beta discovered that software solutions were often sold at revenues five times higher than domestic custom development services. They also found that many clients were eager to quickly improve their sales through e-commerce.

After a prolonged period of collective interpretation, Beta formulated a strategic consensus in 2013 to specialize in e-commerce solutions and products.

In our routine Wednesday afternoon meeting, we (BU managers) discussed [the] feasibility of the e-commerce solution (e-DRP) as a direction to pursue. After more than six months' discussion, we finally chose to be an e-commerce solution provider. (BU manager Cao)

Developing resilient IT portfolios. Beta internalized new domain knowledge and experiences learned while conducting projects to improve their IT products and solutions portfolio. Resilient IT portfolios are characterized by the flexible assembly of diverse IT services to meet different requirements. *Developing resilient IT portfolios* is the process of differentiating, optimizing, and integrating alternative IT services to form a loose-coupled service portfolio around a clear strategic goal.

IT service differentiation refers to different BUs focused on different domains or developed different IT services. Each BU had its own sub-domains when undertaking domestic custom development projects. For example, Cao and her team focused on B2C (business-to-consumer) and B2B (business-to-business) platforms, while Zhong's team worked on channel management and developed products such as online community business circles. Another BU, managed by Li, focused on order management, warehouse and logistics management, and customer management. This approach helps Beta expand its product portfolio and incorporate the latest digital technologies to meet new environmental requirements.

Moreover, Beta encouraged BUs to develop various products and solutions based on their market insights and capabilities respectively. For example, Cao proactively established a team in her BU to develop mobile applications in 2014, anticipating the explosive growth of mobile Internet usage. Her team also collaborated with several clients, including a national laundry chain, an art auction house, and a luxury goods sales agent, to develop a mobile online shopping mall. As the mobile Internet grew rapidly in China, the mobile team became increasingly essential to Beta, since most e-commerce projects needed mobile portals. In 2015, as online-to-offline applications became prevalent among numerous traditional companies, Zhong and her team developed an online-to-offline (O2O) omnichannel operation solution. Later, another team created a big data analysis platform, which was initially based on the business requirements of a large sportswear company. As the COO Ting said, "BU managers can decide which market they wants to target."

IT service optimization refers to renewing and upgrading functions of IT services or reducing costs of IT services. Beta's IT products and solutions were continuously refined and optimized as they were adopted by multiple clients. As the e-DRP Project Manager (PM B1) commented, "once a client implemented the product, we gained new experience, which we then use to upgrade and enhance the product." For example, at the request of a laundry chain store, Beta developed an e-commerce solution with dual portals for PCs (personal computers) and mobile phones. In the next version, Beta upgraded the e-commerce solution with an Android application based on this project. Beta assigned a professional product team to improve e-DRP continuously based on ongoing adoptions and the latest technological trends. For example, they renewed the latest version of applications in response to iOS system upgrades. The upgraded version of e-DRP, an O2O omnichannel operation solution, received numerous awards, including the Best e-Commerce Solution Award by the Ministry of Industry and Information Technology (MIIT) and China InfoWorld in 2015 and the China Software Gold Award by the China Software Industry Association (CSIA) in 2016.

IT service integration involves the combination of different IT services at the company level around a clear goal. For instance, Beta reached a

consensus on conducting e-commerce projects to expand the domestic market after collective interpretation of business domains, and gradually e-commerce related projects accounted for 40 % of the business in the first quarter of 2015, which facilitated the accumulation of e-commerce domain knowledge. As BU manager Cao noted,

The direction of our entire market is shifting to e-commerce, and our domestic sales staff are basically in this e-commerce field. Moreover, it is also obvious that whenever a case is mentioned, it essentially involves buying and selling things on the Internet.

Due to Beta's consistent focus on e-commerce projects, diverse IT services from different BUs were integrated into a comprehensive e-commerce product portfolio, spanning front-end user interfaces to back-end data processing. At the beginning of 2015, Beta introduced a new concept of EC360—an enterprise Internet ecosystem solution that integrated all of Beta's products, including digital marketing, supply chain reconstructing, industrial Internet, and SOA services—to facilitate clients' digital transformation.

Developing resilient IT portfolios shows that every BU specialized in one or more system elements, and they collaborated based on client requirements. For example, two BUs jointly developed e-commerce solutions for a national laundry chain, integrating mobile applications and store management systems. Cao once proudly talked about the mobile Internet team that she established in her BU: "Whenever the company engages in mobile application development, they will need my team."

Developing resilient IT portfolios and *collective interpretation of business domains* are mutually reinforcing. The latter provides business and technology domain knowledge for the former and a clear strategic consensus for combining different IT services. The former provides clear targets for acquiring, integrating, and assessing domain knowledge.

4.3.3. Solution and experience demonstration to strengthen the feedback

The product portfolio and capabilities strengthened positive market feedback by ability demonstration that helped gain more projects.

Solution demonstration. *Solution demonstration* by vendors refers to showing potential clients their ideas and specific solutions based on clients' business requirements, while demonstrating their understanding of clients' business. Beta established a cross-functional pre-sales team consisting of software engineers, business consultants, and salespeople to interact with clients to propose satisfactory solutions tailored to their requirements. This approach helped to improve the success rate of sales.

When we communicated with Firm H (a home appliance manufacturer) before the contract, they did not know how to do it. We not only gave them software solutions but also made corresponding business plans. (PM B2)

Moreover, the cross-functional team also collaborated with consulting companies when bidding on major custom development projects. Management consultants offered solutions from business perspectives, including management, operation, strategy suggestions, and business model design, while Beta focused on the technical aspects, primarily e-commerce-related systems, such as requirements analysis, system design, and implementation. Zhong, one of the BU managers, described her collaboration with a consulting company,

The consulting company elaborated from a relatively high perspective, including how the enterprise should operate the company, how to raise money, the future revenue expectation after the e-commerce operation, and so on. We gave the customer a suggestion on how IT could improve their operation, and what expected effect could be achieved.

Last, Beta was able to create demos quickly for clients based on the existing product to showcase the anticipated outcomes and detailed requirements. For example, when Beta competed for the custom development project from a top-ten appliance manufacturer, it developed a demo based on the existing e-DRP product, which pleasantly surprised the client's CEO.

Experience demonstration. *Experience demonstration* refers to

demonstrating previous successful cases and experiences to win clients' trust. Beta introduced its successful cases to potential clients, highlighting the actual value brought to previous clients, which increased clients' confidence in Beta.

The CEO of Firm H once said, "I want a secondhand product. I hoped that the product had already been used by other companies, and (in this case) I would feel safer." (PM B2)

In conclusion, Beta formed an IT products/solutions portfolio and built new capabilities using knowledge internalization (i.e., IT services optimization, differentiation, integration, and domain knowledge acquisition and integration) by delivering custom development services to diverse clients. As one industry expert said, "A customized system needs numerous cases for polishing. Only after refining can it become a product suitable for more enterprises." In return, the products/solutions portfolio and new capabilities boosted Beta's ability to gain new clients through ability demonstration activities, such as solution demonstration and experience demonstration, strengthening positive market feedback. Hence, Beta formed a new stable state between itself and the environment, as illustrated in Fig. 4.

4.4. Alpha's failure in establishing a resilient feedback loop: client-specific positive feedback and original capabilities sustained the status quo

Alpha's initial exploration of providing IT products and domestic custom development services (discussed in 4.2) generated positive client-specific feedback (see Table 7). This prompted Alpha's top managers to shift their focus to a few critical customers and further improve their existing capabilities and resources, which in turn strengthened positive client-specific feedback. We will discuss this loop in detail in the following section.

4.4.1. Positive client-specific feedback generated by three alternative IT services

Alpha expected the Japanese market to recover soon while exploring the domestic market. Although its biggest Japanese client showed some signs of recovery in 2010 after significantly reducing orders in 2009,

Table 7

External positive feedback generated by IT services of Alpha.

IT services	Positive client-specific feedback	Positive market feedback
Offshore IT outsourcing	+: Market demand declined, as clients only commissioned urgent and challenging tasks. Moreover, profit margins were impacted by fluctuations in the RMB/Yen exchange rate: the Yen's depreciation triggered a substantial decline in profitability. "The profit rate of Japanese services is low because of exchange rates, so our goal now is to break even." (Partner Fu, in 2010)	N/A: No new client. "The Japanese market remained stagnant. That is, we do not have new clients, but only existing ones." (Partner Fu)
Domestic custom development	++: InTr adopted e-Commerce strategy as a differentiated approach, including building an e-Commerce platform supported by a B2B distribution platform, a B2C website, and a remote call center in different places (Annual Reports of InTr from 2011 to 2013), which has continuously generated IT services orders for Alpha over consecutive years.	N/A: InTr was a large travel agency, but the experience gained from it could not be applied to other travel agencies. "The work we do for InTr does not apply to others." (PM A2, Alpha) No other potential clients before 2016.
Solutions/Products offering	+: Two national restaurant chains, which consisted of hundreds of small restaurants, did not expand their restaurant networks.	+: The product, sold to two national restaurant chains and other small eateries, remained unprofitable and was eventually discontinued in 2018.
Overall	+++	+

Note: +++/++ refers to strong external positive feedback.
+ refers to weak external positive feedback.
N/A refers to that no feedback is found.

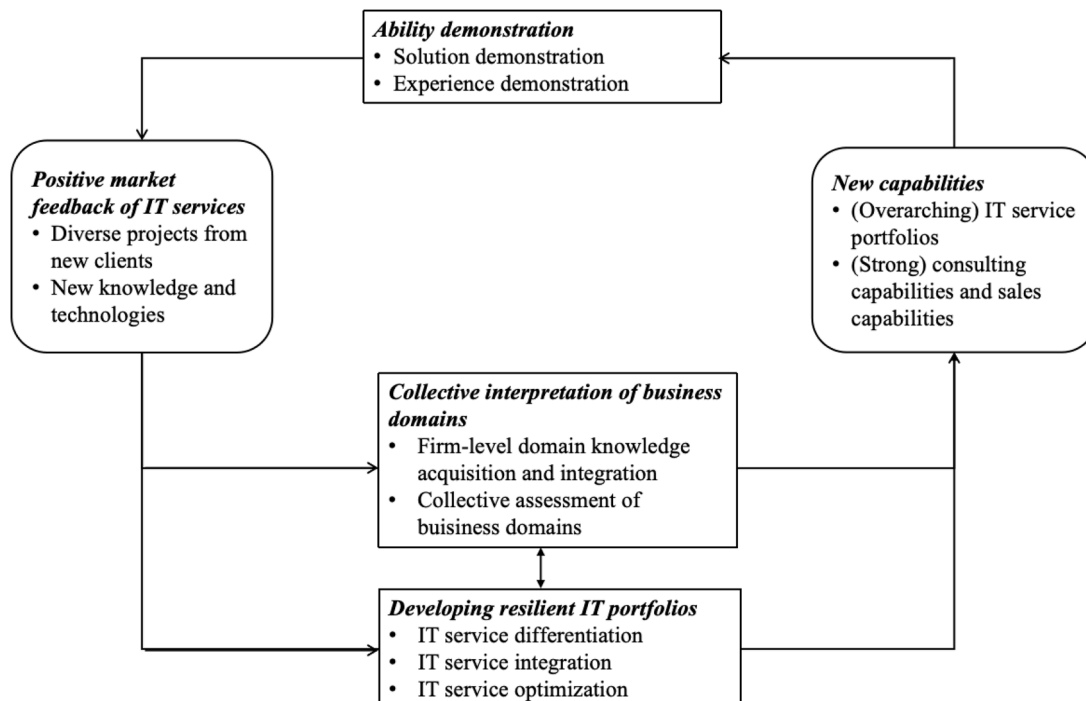


Fig. 4. Beta's resilience feedback loop in the domestic market.

Alpha only received “urgent and difficult tasks” with very thin profit margins (Partner Fu), making the offshore custom development services less attractive than before. To make things worse, the drastic appreciation of the Chinese RMB against the Japanese yen further squeezed the profit margins of Alpha.⁶ As a result, the positive client-specific feedback that had been established in the Japanese market weakened, and positive market feedback vanished entirely (see Table 7). Despite that pessimistic outlook, Alpha preserved the offshore team for the Japanese client to maintain the firm’s survival.

Meanwhile, Alpha’s other two initial explorations were in the same situation (see Table 7). First, the business of IT products generated positive client-specific feedback and weak positive market feedback. In 2009, Alpha aggressively ventured into the domestic market, leveraging its experience with InTr to develop software products like tourism industry e-commerce solutions. However, these products saw minimal sales due to a lack of market demand: medium-sized travel agencies develop B2B transaction systems themselves, while small ones could not afford such systems.

In 2010, Alpha piloted a new business model to overcome its lack of marketing and sales capabilities. It partnered with a dealer to develop an order processing system for Chinese restaurant chains. Alpha was responsible for the development while the dealer conducted requirements analysis and sales. In just one year, this order-processing product was sold to two national restaurant chains, including over 100 small restaurants. Zhang and Fu had anticipated the product would become a primary sources of profit by 2012, and they even hired a salesperson to support growth. However, sales declined in 2013, leading to the salesperson’s dismissal. Furthermore, the product required continuous maintenance and upgrades, but its income was insufficient to cover the subsequent investment.

Fu became pessimistic about the product’s future. He said, “*This product is very difficult to make a profit, and we can hardly get the development cost back.*” As a result, Alpha only allocated two developers to the maintenance and upgrade of the product based on the dealer’s requirements. Subsequently, the maintenance tasks turned into a custom development project, and the dealer became Alpha’s client. This product only became profitable in 2015 but was pulled from the market in 2018 following the dealer’s bankruptcy. In the aftermath, Alpha decided to cease all new product development.

Second, Alpha’s domestic custom development service received positive client-specific feedback but lacked positive market feedback. Despite struggling to expand domestic custom development services, Alpha only acquired a few new clients with small-scale projects such as an e-commerce website, a short message system, a membership management system, and an after-sales service system for various companies. Notably, all these small projects were one-time collaborations.

Fortunately, custom development projects from InTr became Alpha’s primary source of profit after 2010. Alpha built an e-commerce website for InTr in 2005, then developed a B2B travel operation system, and continued providing maintenance services thereafter. When InTr went public in 2010, it expanded its domestic travel agency network to smaller cities, and implemented a differentiation strategy by building an integrated e-commerce platform. This platform combined a B2B travel-operation platform, a B2C website, and a remote call center in different places. This active creation of demand for IT services led to several development and maintenance contracts for Alpha related to InTr’s e-commerce platform since 2010. Although InTr projects generated strong positive client-specific feedback, they did not help Alpha penetrate the travel agencies market, making it difficult to generate positive market feedback. A project manager (PM A2) of the mentioned InTr project explained,

InTr is one of the biggest travel agencies in the country and holds a dominant position in the Chinese tourism industry. Other travel agencies differ from InTr in terms of their services and operations. All the solutions we developed were tailored to InTr’s unique needs, and we found that they were not transferable to other companies. We once attempted to collaborate with ChiTr (pseudonym, another large travel agency), but they declined when they learned about our involvement in InTr’s project.

Fu did not turn his attention to the domestic market until 2013. After that, he attempted to develop a second or third client similar to InTr or intermediaries that needed IS development but lacked the technical capabilities, regardless of their industry or the types of systems they used. However, Alpha only secured one potential big client in the energy industry in 2013, and they failed to form a substantial large-scale collaboration until the middle of 2015. In 2016, Alpha became a wholly owned subsidiary of a local consulting firm in the finance industry, helping the company develop a localized commodity trading and risk management system.

4.4.2. Developing isolated IT portfolios and individual interpretation of business domains to strengthen existing capabilities

Fu had repeatedly stated that Alpha’s primary focus was the domestic market after 2013, however, more than half of the employees remained engaged in Japanese projects. Since 2013, Alpha had two main businesses: offshoring IT outsourcing development and domestic custom development. These businesses generated positive client-specific feedback (refer to Table 7), following the same pattern of providing custom development and maintenance services to a few large clients/partners. Driven by positive client-specific feedback, Alpha developed isolated IT portfolios and undertook individual interpretation, which strengthened the existing client-specific and software development capabilities.

Individual interpretation of business domains. Individual interpretation of business domains refers to the concept that managers individually acquire external information and domain knowledge to analyze market demands of new business domains without engaging in firm-level discussions about such analyses. Zhang, who was in charge of Alpha’s product development before 2013, learned about the order processing system’s requirements from various sources such as salespeople, dealers, restaurant customers, and personal friends. However, other BU managers, such as Fu, did not mention how they learned about new environments.

Before 2013, there was little information and knowledge integration among managers. Alpha’s executive meetings, attended by BU managers, mainly discussed internal management issues like bonus allocations once per month. Meanwhile, meetings attended by top managers, BU leaders, and stakeholders were held only two to three times per year. Thus, this reliance on individual interpretation led to a siloed working environment, preventing managers from reaching a strategic consensus. For example, Zhang advocated that IT products were the future of Alpha, while other managers prioritized custom development services. Due to this lack of consensus, Alpha did not have a clear and consistent goal.

Developing isolated IT portfolios. Developing isolated IT portfolios refers to the concept that IT service vendors create, upgrade, deploy and provide IT services that are independent of each other, including *IT service differentiation and optimization*, as described below.

Alpha implemented *IT service differentiation* through three independent BUs: The first BU provided services to its biggest Japanese client, the second focused on InTr, and the third one developed software products for other Japanese clients. Each BU had dedicated employees and served specific clients for a long time, accumulating client-specific knowledge, such as operational workflows and knowledge about IS and architectures. Furthermore, each BU also conducted specialized technical and business training tailored to its clients or operations. For example, the domestic BU trained new hires on InTr’s B2B systems and imparted travel industry expertise while streamlining project

⁶ For example, 100 Yen would be equivalent to about 8 RMB in 2008, but in June 2013, down to about 6 RMB only, depreciated over 20%, https://srh.bnkofchina.com/search/whpj/search_cn.jsp.

management practices to reduce costs. In contrast, the Japanese BU emphasized Japanese language training, skills for programming languages, and software development norms. This targeted training system improved learning efficiency.

The Japanese BU's training focuses on Japanese language and software development processes instead of business knowledge. The domestic BU's training involves software development processes, original system architecture, and related domain knowledge. (Fu)

Although IT service differentiation contributed to client-specific capabilities, it simultaneously impeded the development of consulting and sales skills required for exploring the domestic market. First, positive client-specific feedback led to the accumulation of homogeneous knowledge to the firm, which proved irrelevant to acquiring domain knowledge [30,31]. For example, InTr's B2B platform was built specifically for InTr, and the subsequent maintenance projects were customized to this platform. Knowledge and experience derived from collaborating with a specific client on a dedicated system were inherently difficult to transfer to other clients. A project manager (PM A2) who took in charge of InTr projects commented that the experiences gained from InTr may not be applicable to other travel agencies.

Second, Alpha had three IT service projects in different industries, including tourism, software, and catering. However, these projects lacked common domain knowledge. Moreover, Alpha did not provide adequate formal and informal training to support the internal communication, integration, and transfer of such knowledge. Therefore, the absence of consolidated domain expertise made it challenging to cultivate the consulting capabilities required for domestic market penetration.

Alpha implemented *IT services optimization* to improve software development capabilities. For example, Alpha created a software development platform for internal use based on years of advanced software engineering experience accumulated from working with Japanese software firms. Alpha then applied this tool to the implementation of domestic services. As described by a project manager,

We have developed a software development platform that can boost productivity by at least 30 % to 40 %. This platform is applicable to any other organization and has been used to create our order processing product. The more you use this platform, the more stable it becomes. (PM A3, Alpha)

Moreover, Alpha also developed an automatic testing tool based on the experience gained from collaborating with Japanese clients. The tool identifies which programs have been tested and which remain untested. Its application in domestic projects has reduced testing time and enhanced development quality, particularly for new employees engaged in coding tasks.

The interaction between individual interpretations of business domains and the development of isolated IT portfolios has created a vicious cycle: the latter undermines cross-project communication, which in turn reinforces the former. As a result, IT service divergence emerges from conflicting perspectives rooted in isolated individual interpretations of external environments.

4.4.3. Experience demonstration to strengthen positive client-specific feedback

Alpha's client-specific and software development capabilities were significantly strengthened through working consistently with a few clients, which, in turn, resulted in a consistent flow of orders from specific clients. Our data show that *experience demonstration*, that is, showcasing successful experiences with previous clients, strengthened positive client-specific feedback. On the other hand, the lack of solution demonstration impeded Alpha's ability to gain new diverse clients.

Conducting *experience demonstrations* helps Alpha secure more projects from existing clients, forming deep but narrow client relationship. Alpha's project managers and BU heads were highly confident in their

capabilities, which guaranteed their delivery quality and long-term collaborations with clients.

We have been developing and maintaining InTr's B2B platform for over ten years. Although InTr has acquired multiple companies since 2010, necessitating a transformation of its B2B trading system, our recent business engagement with it has remained robust. In fact, we have become a core IT service vendor for InTr. (Fu)

Fu emphasized the importance of quality excellence and on-time delivery to his employees, believing that a successful project could generate 20 new ones and that clients would only continue to cooperate if projects were executed excellently. However, despite demonstrating past experience, Alpha failed to attract new clients because it couldn't showcase solutions that required consulting and sales capabilities—capabilities that went beyond mere software development. Unfortunately, the lack of these two capabilities has always been a major obstacle for Alpha to explore the domestic market. Fu admitted that they had a difficult time trying to understand clients' business requirements,

Sometimes, technology may not be the primary driver in market exploration. Due to our technical background, our team may sometimes use technical jargon when communicating with clients, leading to comprehension gaps. Similarly, clients may also use business terms that are hard for us to understand.

In turn, the dearth of new projects impeded Alpha's development of new capabilities for exploring new services. For example, Alpha once tried to adopt new technologies and applications, such as big data, the mobile Internet, and digital market platforms, but it made no progress due to a lack of implementation opportunities. This has created a negative feedback loop for Alpha, leading them to become more entrenched in their existing pattern of providing long-term services to a small number of clients (see Fig. 5).

5. Discussion

5.1. IT service vendors' resilience feedback loop

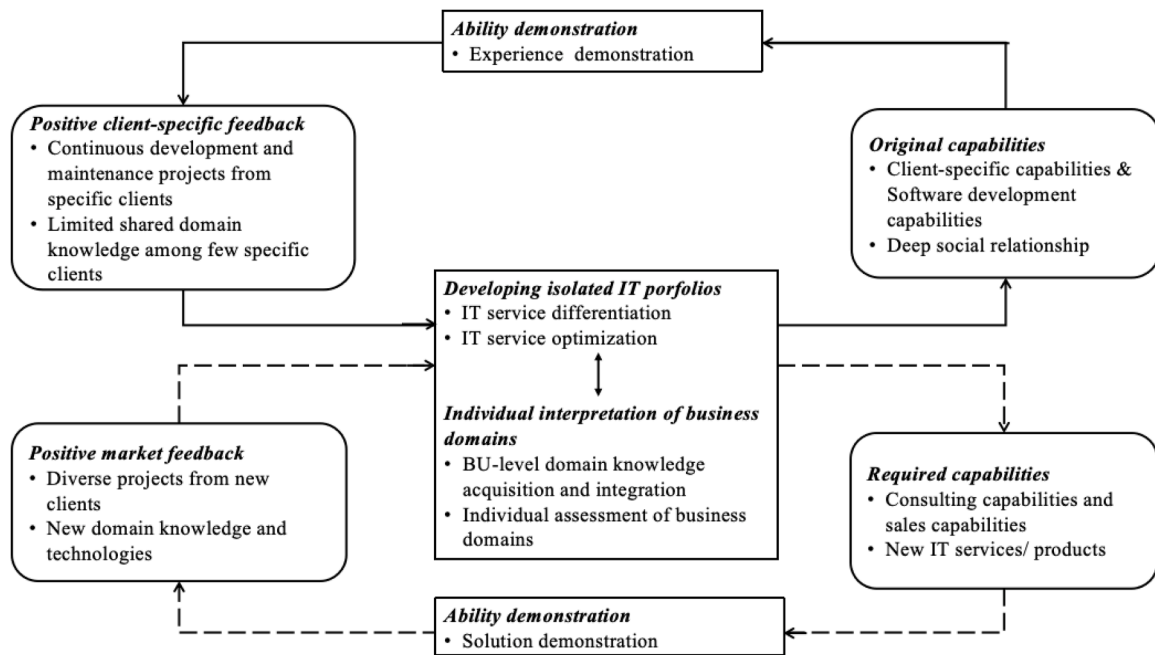
Our research shows that the resilience feedback loop between external environment and organizations involves an inside-out process and an outside-in process. Organizational actions like IT service adaptation, environmental interpretation, and ability demonstration facilitate these processes (see Fig. 6).

5.1.1. Outside-in process: from external positive feedback to internal capabilities

The outside-in process refers to how IT service vendors access, analyze, and integrate external resources—such as knowledge and information—into their internal operations to build or reinforce capabilities. This process begins with external positive feedback, including both market and client-specific feedback, which emerges from IT service vendors' initial responses to major exogenous shocks. Such feedback reflects the strength and nature of resource linkages between an organization and its external business environment [94].

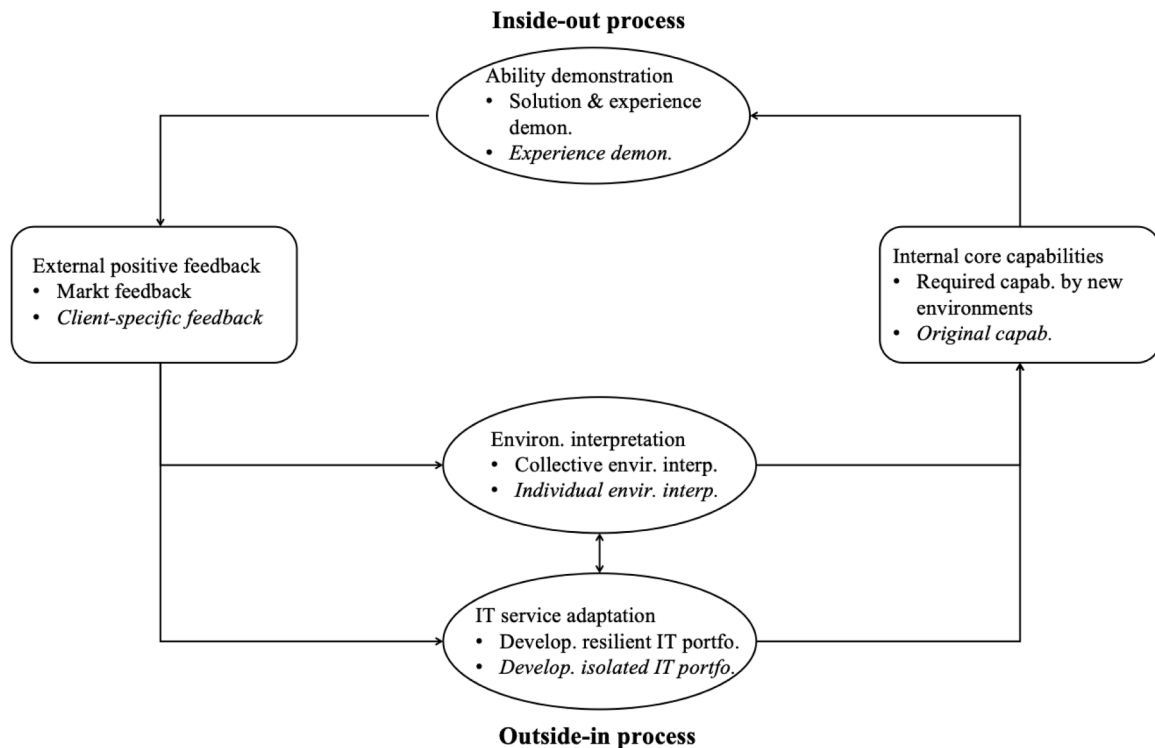
Specifically, positive market feedback attracts diverse projects, business requirements, and domain knowledge, enabling IT service vendors to build new capabilities. In contrast, positive client-specific feedback introduces more homogeneous business requirements and domain knowledge from existing clients, reinforcing existing capabilities. The differing outcomes stem from two critical actions within the outside-in process: environmental interpretation and IT service adaptation.

5.1.1.1. Environmental interpretation. Environmental interpretation entails assessing and interpreting environment risks and opportunities [15]. Results of interpretation shapes IT service vendors' decisions on



Note: solid arrows represent promoting effects, dashed arrows represent hindering effects

Fig. 5. Alpha's feedback loop in the domestic market.



Note: Normal font depicts the successful process of building resilience;
Italic font refers to a failure process of building resilience

Fig. 6. The resilience feedback loop of IT service vendors.

how to respond to exogenous shocks [76]. We find two kinds of environmental interpretation: collective and individual interpretation of business domains. They lead to divergent understandings of the challenges and opportunities posed by major exogenous shocks.

Positive market feedback—Collective interpretation—Building new

capabilities. Collective interpretation refers to IT service vendor managers acquiring external information and knowledge independently, then jointly analyzing and integrating it. This process aligns with the activity of loosely coupled organizational modularity [98,99].

Positive market feedback provides diverse knowledge, helping

managers who conduct collective interpretation in loosely coupled IT service vendors better understand new environments [99] and identify business opportunities [32]. This enhances vendors' confidence in engaging in proactive measures to respond to external shocks [54], such as developing new capabilities as highlighted in our research. In addition, managers of IT service vendors enrich domain knowledge during joint acquisition and integration of external knowledge from various clients, contributing to building consulting capabilities [100].

Positive client-specific feedback—Individual interpretation—Reinforcing original capabilities. Individual interpretation refers to the way that IT service vendor managers independently acquire and assess external information and knowledge.

Positive client-specific feedback indicates that these vendors face challenges in exploring new markets following significant external shocks. When relying on positive client feedback, IT service vendors tend to gather similar information and knowledge only from specific clients. When this information is collected, analyzed, and evaluated by managers independently—without firm-wide discussions—conflicts may arise among them. The limited information available, along with differing opinions and prior failures related to exploration, can lead to inertia within IT service vendors [101]. This inertia results in negative responses to external shocks [15], making it difficult for these vendors to identify new business opportunities [102]. Consequently, this situation reinforces the status quo and maintains existing capabilities.

5.1.1.2. IT service adaptation. IT service adaptation involves responding to emerging demands arising from exogenous shocks by learning from external resources and knowledge to design and develop IT services [21]. This includes developing resilient IT portfolios and isolated IT portfolios. They contribute to capability building.

Positive market feedback—Developing resilient IT portfolios—Building new capabilities. IT service vendors develop resilient IT portfolios by differentiating, optimizing and integrating IT service products and solutions while learning from and incorporating external domain knowledge. Developing resilient IT portfolios helps IT service vendors build new capabilities.

First, resilient IT portfolios are associated with technical modularity, which refers to the practice of using loosely coupled system functions. This resilience enhances vendors' efficiency (via optimization) and fosters innovation (through differentiation and integration), allowing them to adapt to new technologies and meet the diverse demands of clients [23], ultimately enhancing their capabilities.

Second, as IT service vendors consistently optimize and integrate their service products [31], managers and employees accumulate and enrich their domain knowledge, which aids in building the consulting capabilities needed to address the requirements of new clients [30,100].

Positive client-specific feedback—Developing isolated IT portfolios—Reinforcing original capabilities. Developing isolated IT portfolios primarily focuses on optimizing and enhancing the efficiency of existing, independent IT services without leveraging synergies. While this approach may enable organizations to “walk on two legs,” it can also result in a lock-in effect, whereby firms become constrained by their established businesses and capabilities due to positive client-specific feedback (e.g., [103]). This inertia can impede the development of new capabilities. Furthermore, IT service vendors will strengthen their core competencies, such as software development capabilities, through the continuous optimization of IT service products.

In the outside-in process, the interplay between environmental interpretation and IT service adaptation shapes internal capability development. Decisions made based on environmental interpretation establish clear strategic objectives for IT service adaptation. For instance, when an IT service vendor chooses to enter a new market following exogenous shocks, they are likely to develop resilient IT portfolios that accommodate diverse client demands. Simultaneously, IT service adaptation necessitates the acquisition and integration of

external knowledge and information (i.e., environmental interpretation). For example, developing resilient IT portfolios requires extensive domain knowledge to design a loosely coupled portfolio structure.

5.1.2. Inside-out process: from internal capabilities to external positive feedback

The inside-out process depicts how IT service vendors change their resource environment by influencing clients' collaboration decisions. Our data indicate that ability demonstration facilitates this inside-out process.

5.1.2.1. Ability demonstration. Ability demonstration refers to the activities used by IT service vendors to showcase their operational capabilities to potential clients. This practice acts as a reputation signal, allowing potential clients to evaluate and choose vendors, which further changes IT service vendors' environmental conditions [104]. It also aids IT service providers in gaining the trust of potential clients and fostering collaboration [3,5,104,105]. We identify two types of ability demonstrations: solution demonstration and experience demonstration. However, the effects of these two types on external positive feedback may differ.

Building new required capabilities—Solution demonstration—Positive market feedback. IT service vendors that develop new consulting capabilities and resilient IT portfolios can conduct the solution demonstration to showcase their understanding of the project and its potential for successful completion. For example, by providing a demonstration sample to a potential client, the IT service vendor can communicate its comprehension of the project's requirements clearly, allowing the client to assess whether the vendor possesses the necessary expertise to execute the project. Additionally, the demonstration provides an opportunity for clients to identify areas for improvement. Consequently, solution demonstration is effective in building trust and fostering collaboration with multiple potential clients, ultimately enhancing positive market feedback and broadening their environmental conditions.

Leveraging original capabilities—Experience demonstration—Positive client-specific feedback. Experience demonstration refers to showcasing successful past IT projects to clients. As new markets emerge, vendors may need to develop new capabilities to meet potential clients' demands. Therefore, demonstrating experience in a new market may not necessarily help IT service vendors gain the trust of new clients. However, it can strengthen trust with existing clients and enhance collaboration [5], ultimately reinforcing positive client-specific feedback.

Overall, the outside-in and inside-out processes together form the resilience feedback loop. Through the inside-out process, IT service vendors can leverage newly developed capabilities to demonstrate their abilities, thereby gaining trust and fostering collaboration with a broader range of potential clients (i.e., positive market feedback). In contrast, the outside-in process allows vendors to acquire and integrate external resources, which are essential for developing the new capabilities required for success. These two processes mutually reinforce each other, enabling IT service vendors to adapt and recover from exogenous shocks, ultimately reaching a new equilibrium and enhancing organizational resilience.

5.2. Theoretical contributions

Our research investigates IT service vendors' organizational resilience and makes three theoretical contributions. First, we expand on the existing research on resilience in IS literature to include a more complex situation: the resilience of IT service vendors. Whereas previous studies emphasize that technical modularity can enhance the resilience of IS, this research investigates IT service vendors' complex process of building resilience and finds that IT service vendors should adopt both organizational modularity and technical modularity to enhance

resilience, echoing the findings of Tim et al. [36]. Furthermore, while previous research has suggested that resilient IS help IT service clients build resilience by changing their own routines and practices [29,50], this study reveals that resilient IS, such as a loosely coupled IT portfolio, enables vendors to attract more diverse clients, thus strengthening their own resilience. This research introduces a new research agenda in IS resilience literature by distinguishing the resilience of IT service vendors from that of IS and IT service clients.

Second, this study combines the research of risk management with that of organizational resilience, enriching the research on IT service vendor resilience. We illustrate that developing more comprehensive capabilities, such as consulting capabilities, sales capabilities, and portfolio services, rather than merely strengthening technical capabilities, is essential for IT service vendors to build resilience and bounce forward from exogenous shocks. Furthermore, we find that increasing the breadth of connection with clients rather than the depth can better promote resilience. Specifically, our research clarifies the roles of different types of client relationships on IT service vendors' organizational resilience. In contrast to previous studies, which proposed that IT service vendors ought to center on establishing robust relationships with individual clients to mitigate project risks (e.g., [66]) and facilitate recovery from external shocks (e.g., [5]), our data indicate that this strategy might impede IT service vendors from adapting to new environments over the long term. Instead, we find that having broad client relationships can be more effective in responding to external shocks. Additionally, we put forward specific practices that can help build new capabilities and broad social relationships, including developing resilient IT portfolios, reaching collective interpretation of business domains, and demonstrating solution and experience, which offer preliminary insights for IT service vendor resilience research.

Third, our research contributes to the literature on organizational resilience from a process perspective. Organizational resilience can be developed through a feedback loop of interactions between the environment and the organization [15]. We refine this process and divide it into two sub-processes: the outside-in and inside-out processes. In the outside-in process, organizations continuously access, analyze, and integrate external resources to build new required capabilities through environmental interpretation and IT service adaptation. These two actions correspond to environmental interpretation and learning conceptualized in the literature [15]. In the inside-out process, organizations demonstrate their new capabilities and resources through ability demonstration to enhance positive feedback from the environment. This classification helps to clearly depict the feedback loop between the environment and organizations since extant research of organizational resilience has primarily focused on how firms exploit external resources (i.e., the outside-in process) while ignoring how firms influence the environment (i.e., the inside-out process). Additionally, this framework identifies "ability demonstration" as a novel action that serves as a reputation signal [104] and constitutes the resilience feedback loop when combined with learning and environmental interpretation. By contrasting the critical actions within the resilience feedback loop with those that fail to establish it, we elucidate the underlying reasons for why some organizations struggle to build resilience.

5.3. Managerial implications

Our research has two managerial implications. First, in the current digital era, new digital technologies are emerging at an unprecedented pace, which amplifies the need for IT service vendors to transform. Our research results indicate that establishing a comprehensive IT portfolio that includes different businesses can help organizations resist external shocks. IT service vendors could design resilient IT portfolios by encouraging employees to update existing digital services/products and build new ones based on collaborations with clients. Moreover, vendors should have a clear and consistent goal when developing new digital products/services, since a well-defined and sustained objective

facilitates the combination of products and provides guidance for managers to build new products/services. Formal information and knowledge-sharing meetings throughout the firm can also facilitate this process.

Second, our research offers actionable insights for vendors in other industries that have encountered exogenous shocks. During tough times, vendors may be inclined to offer price concessions to their key clients to secure long-term partnerships and enhance competitiveness. However, this approach is not always effective, and blindly selecting customers could lead to even greater losses. Therefore, we advise managers to exercise prudence in partner selection. An ideal partner is one who can provide external resources to assist firms in developing market-requisite new capabilities.

5.4. Limitations and future research

This research has three major limitations. First, our data collection has some limitations that could potentially impact the quality of this research. We collected retrospective data on Beta from 2014 to 2015, a period when Beta had already recovered from the financial crisis of 2008 and developed resilience. Retrospective reports may lead to errors and omission of key incidents [106]. To mitigate this, we adopted triangulation among informants and Beta's website. When discrepancies arose in interviewees' accounts, we gave greater weight to the interviewee responsible for the event because he/she had a deeper memory and a lower probability of error [106]. Although we adopted triangulation, the validity of findings remains at risk, especially for such process research. Moreover, we were unable to obtain interview data after 2015 to track Beta's domestic market progress. Although we monitored Beta's websites and news on the Internet regarding its development after 2015, it is still a significant limitation of this research.

Second, we did not make a distinction between the various forms of feedback that vendors receive from their environment. Feedback can take many forms, including order feedback, knowledge feedback, and performance feedback, and each may play a different role in building resilience for IT service vendors. Future research could systematically analyze these feedback types and their differential effects on organizational resilience building. In addition, we did not examine the impacts of external negative feedback on IT service vendors' resilience. Future research could explore how external negative feedback hinders IT service vendors from building resilience.

Third, our study identified a resilience feedback loop of IT service vendors in response to major exogenous shocks. Hence, new concepts developed in this research—such as the development of resilient/isolated IT portfolios, solution demonstration, and experience demonstration—may not generalize to other sectors (e.g., manufacturing). Future research is needed to validate these findings in different industries. In addition, our sample included Chinese IT service vendors operating primarily in the domestic market. Chinese IT service clients exhibit unique characteristics, such as a focus on personal relationships [5], vendors' consulting capabilities [87], and the demand for new digital technologies, all of which place specific requirements on vendors. Due to the uniqueness of this context, the generalizability of our findings may be limited. We advocate for future research to explore IT service industries in other countries to validate these findings.

6. Conclusions

We conducted a case study to compare the responses of two different IT service vendors to major external shocks. Through this study, we conceptualized a resilience feedback loop for IT service vendors, comprising two interrelated processes: the outside-in and inside-out processes. The outside-in process is built through a combination of developing resilient IT portfolios and collective interpretation of business domains, while the inside-out process is formed through ability demonstration. Our research findings advance the theoretical

understanding of resilience in both information systems (IS) literature and organizational resilience research.

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CRedit authorship contribution statement

Fang Su: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ji-Ye Mao:** Writing – review & editing, Project

administration, Investigation, Funding acquisition, Conceptualization. **Bing Wang:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests, personal relationships, or affiliations that could influence or be perceived to influence the work submitted for publication:

The authors declare no competing financial, professional, or personal interests that could have influenced the outcomes or interpretation of this research.

The funding organizations and affiliated entities listed above had no role in the study design, data collection, analysis, manuscript preparation, or decision to publish.

Appendix

Fig. A1

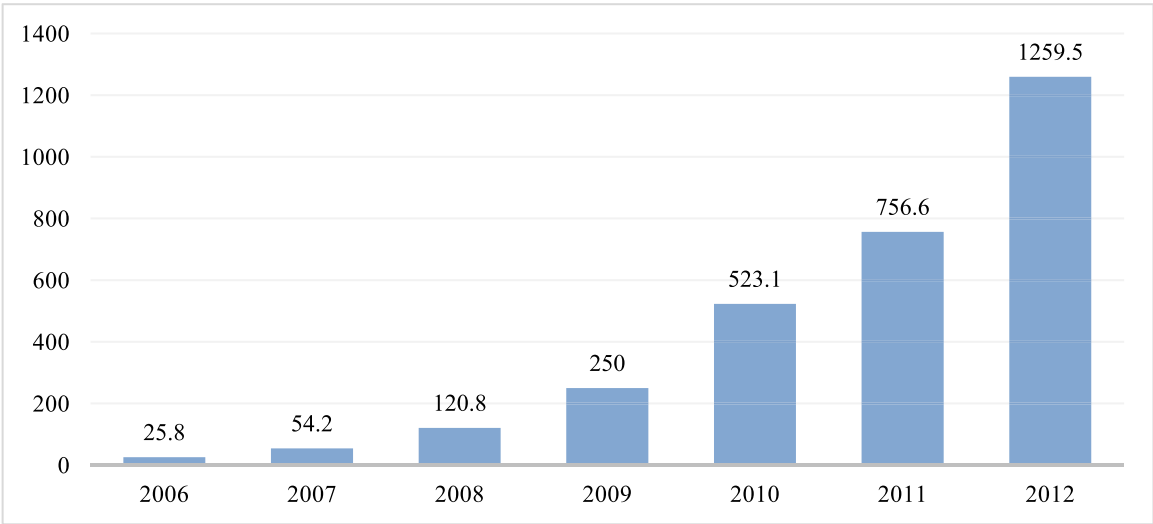


Fig. A1. The amount of Chinese e-commerce from 2006 to 2012 (unit: billion RMB).
Data source: China Internet Network Information Center, *Research Report on China's Online Shopping Market in 2012* (In Chinese), p. 17, <http://www.cnnic.net.cn/hlwfzyj/hlwxbzg/dzswbg/201304/P020130417543965742695.pdf>.

Table A1
Data supporting interpretations of second-order codes.

Dimensions	Themes	Representative quotes
IT service adaptation	IT service differentiation	"Our company is divided into two parts: domestic business and Japanese business, and there is very little communication between them." (PM A1, Alpha) "The two teams (the domestic team and the Japanese team) basically have no staff turnover." (PM A2, Alpha) "Our company's current main business direction is e-commerce, but e-commerce has different domains, such as community e-commerce or business circles, and some B2C, B2B malls, etc. My work is mainly about the community business in e-commerce" (BU manager Zhong, Beta)
	IT service optimization	"We have a prototype product, based on which we do customized development for customers. According to some of their (new customers') requirements, we make corresponding changes or add some new functions. The overall business architecture, including the system structure, has not changed much." (PM B2, Beta) "We have also done some internal development and testing work, including product updates based on the latest updates on the e-commerce websites. For example, we connected the data of Taobao before the official launch of Tmall. But after Tmall came out, we connected the data of Tmall instead of Taobao. Later, Jushita was developed, and all orders must be synchronized from the data of Jushita. We have made a corresponding adjustment to the product." (PM B1, Beta) "We developed a software development platform which improve productivity by at least 30–40 %. This platform can be applied to any firm. Our catering ordering product is developed on this platform. This platform needs to be used many times because the more you use it, the more stable it will be." (Partner Zhang, Alpha)

(continued on next page)

Table A1 (continued)

Dimensions	Themes	Representative quotes
Environmental interpretation	IT service integration	“We have developed tools that we ask newcomers to use to test the code and ensure the quality of the project.” (Partner Zhang, Alpha) “The entire company’s mobile team is in my department. So, they have to work with me as long as the company is involved in mobile development (business).” (BU manager Cao, Beta) “We have also selected a few staff to form a team to continue to improve our products. We gain new insights with each implementation and optimize the product further.” (PM B2, Beta) “The entire market is shifting toward e-commerce, and our domestic sales team is mainly focused on this field. It’s obvious that nowadays, any business case mentioned basically involves online transactions.” (BU manager Cao, Beta)
	(Firm-level or BU-level) Domain knowledge acquisition and integration	“For e-commerce companies like us, we must not only focus on the system but also on how to operate, how IT cooperates with the enterprise operation, and then what expected results can be achieved. These are all things we need to learn.” (BU manager Zhong, Beta) “The training of business domain knowledge mainly focuses on the core business of our entire company, which is e-commerce, such as B2C, B2B2C, C2C, e-DRP, etc.” (BU manager Cao, Beta) “We follow some industry dynamics, and we also refer to the solutions of our competitors. We then learn a lot when working with good partners to develop our own solutions.” (BU manager Zhong, Beta) “I spend more time paying attention to domestic customers, such as meeting customers, seeking clues, and learning about the catering industry.” (Partner Zhang, Alpha)
	(Individual or collective) assessment of business domains	“We have a meeting of middle managers every Wednesday afternoon, which mainly discusses payment issues, company direction, and so on.” (BU manager Cao, Beta) “We know e-commerce solutions are feasible, but what’s next? The e-commerce trend might be temporary, so we need to find the second growth curve quickly. Each manager is thinking about this, and we analyze every completed project to see if it’s worth packaging into a product.” (COO Ting, Beta) “We rarely have routine meetings to discuss a fixed topic. Sometimes, regular meetings may slowly become a routine without any real meaning.” (General Manager Fu, Alpha) “Everyone is responsible for a different BU. For example, if I bring up a new business, not everyone would provide their opinions. At most, I am reminded by them that this business would not be easy to do. I would like to describe the business to them, but they can’t understand it either, because they don’t pay attention to it.” (Partner Zhang, Alpha)
	Ability demonstration	Experience demonstration “We’ve collaborated with Firm RL (a major Japanese software company) for years. It has awarded us annually for our quality... For projects we’re familiar with, RL assigns them to us directly. For complex projects, RL gives them to us if its subsidiaries can’t handle them.” (Partner Zhang, Alpha) “We have a product called e-DRP. Once e-DRP is developed, we can present specific cases to customers and explain the profit margins after implementation, as it’s already a mature product. We’ve found this is indeed a potential market.” (Engineering Director Chao, Beta) Solution demonstration “The sales team first establishes relationships with customers, and my role primarily involves consulting and assisting in solution design. In the early stage, we communicate with clients to identify their business needs, then advise them on how to build the system in alignment with their business models.” (BU manager Zhong, Beta) “When we communicated with Firm H (a home appliance manufacturer) before the contract, they were unsure about how to do it. We provided not only software solutions but also customized business plans, because we have such implementation experience in other bigger home appliance manufacturers.” (PM B2, Beta)

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